

ENV-044V

WIRELESS VERTICAL RUNNING WHEEL

User's Manual



DOC-253
ENV-044V USER'S MANUAL
Rev. 1.1

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MED Associates, Inc.
P.O. Box 319
St. Albans, Vermont 05478
www.med-associates.com

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Table of Contents

Chapter 11

 Overview..... 1

 Technical Specifications 1

 Dimensions..... 1

Chapter 22

 Hardware 2

 ENV-044V Battery Installation 2

Chapter 33

 Operation..... 3

 Setting the Channel Number 3

Chapter 45

 Software 5

Chapter 55

 Cleaning Instructions..... 5

Appendix A.....5

 Contact Information 5

CHAPTER 1

Overview

Med Associates' easy to use light weight aluminum wheels fit on any standard wire topped home cage and require no complex set-up or configuration. These wheels are perfect for long term circadian rhythm studies and allow researchers to quantify running levels and how they correlate with behavioral changes. The wheels are compatible with our Wireless Running Wheel Interface System.

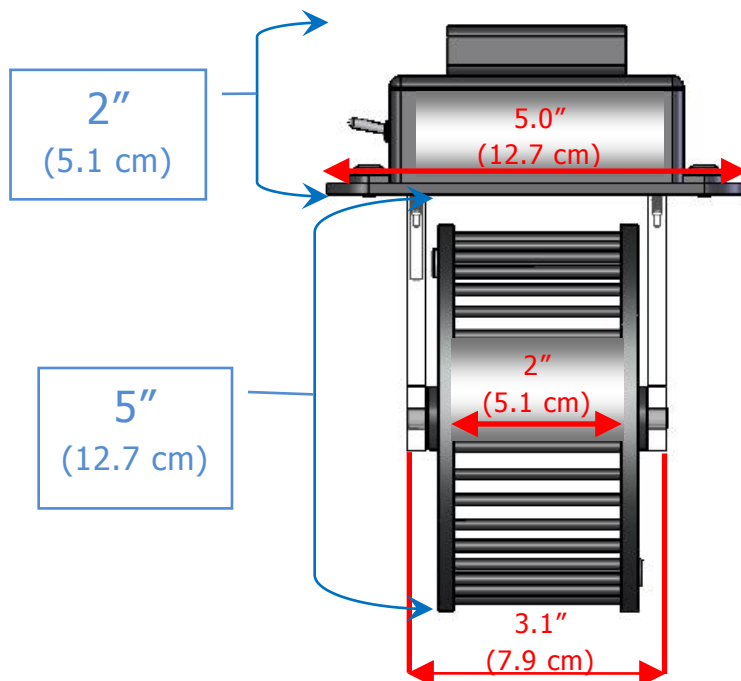
Technical Specifications

Operating Frequency:	2.4 – 2.483 GHz
Range:	~ 50 feet (15 meters)
Batteries:	Three AAA batteries

Dimensions

Total:	5" W x 7.0" H x 4.75" D (12.7 x 17.78 x 12.06 cm)
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Figure 1.1 – ENV-044V Wireless Vertical Running Wheel



CHAPTER 2

Hardware

The ENV-044V Wireless Vertical Running Wheel is used with the DIG-804 USB Interface Hub for quantifying and transmitting running activity. It features a power switch on the left side of the control enclosure, so it is not necessary to disconnect the batteries when resetting the wheel. The power switch is in the ON position (indicated with —) when pulled forward and OFF (indicated with O) when pushed back.

Figure 2.1 – Wireless Vertical Running Wheel shown in Home Cage



Figure 2.2 – DIG-804 USB Interface Hub Front and Back Panel



ENV-044V Battery Installation

1. Remove the cover on top of the battery case by applying pressure and sliding it open.
2. Install 3 AAA batteries.

CHAPTER 3

Operation

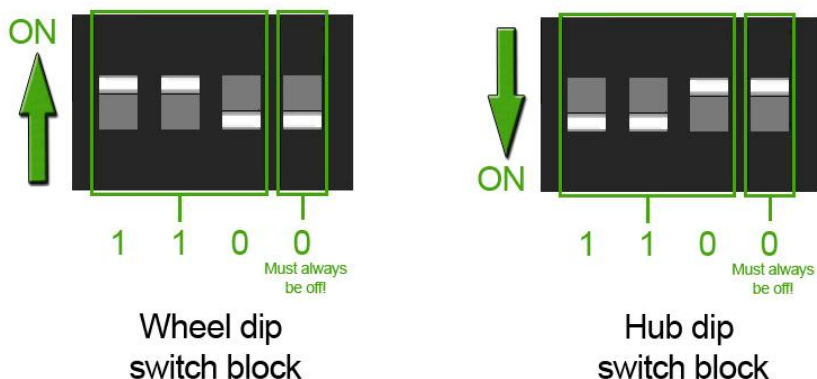
The ENV-044V Wireless Vertical Running Wheels will transmit data, or messages, to the Hub as long as they are connected to battery power and are set to the same channel. The transmission of messages occurs regardless of whether data is being acquired or not. Thus, if the wheels will not be used for data collection for a prolonged period of time it is recommended that the power switch is turned off. Removal of the batteries to prevent damage from corrosion is recommended for extended periods of non-use. If the unit is positioned on the home cage as shown in Figure 2.1 on page 2, then the power switch is ON when pulled forward and OFF when pushed back. More detail is shown in Figure 3.2 on page 4. The messages from the running wheels contain the total number of wheel revolutions since the last message was sent to the Hub.

Setting the Channel Number

NOTE: Be sure that the DC power cable to the DIG-804 is disconnected and the power switched off on ENV-044V before making changes to the settings of the channel selection switches; otherwise the **changes will not be recognized**.

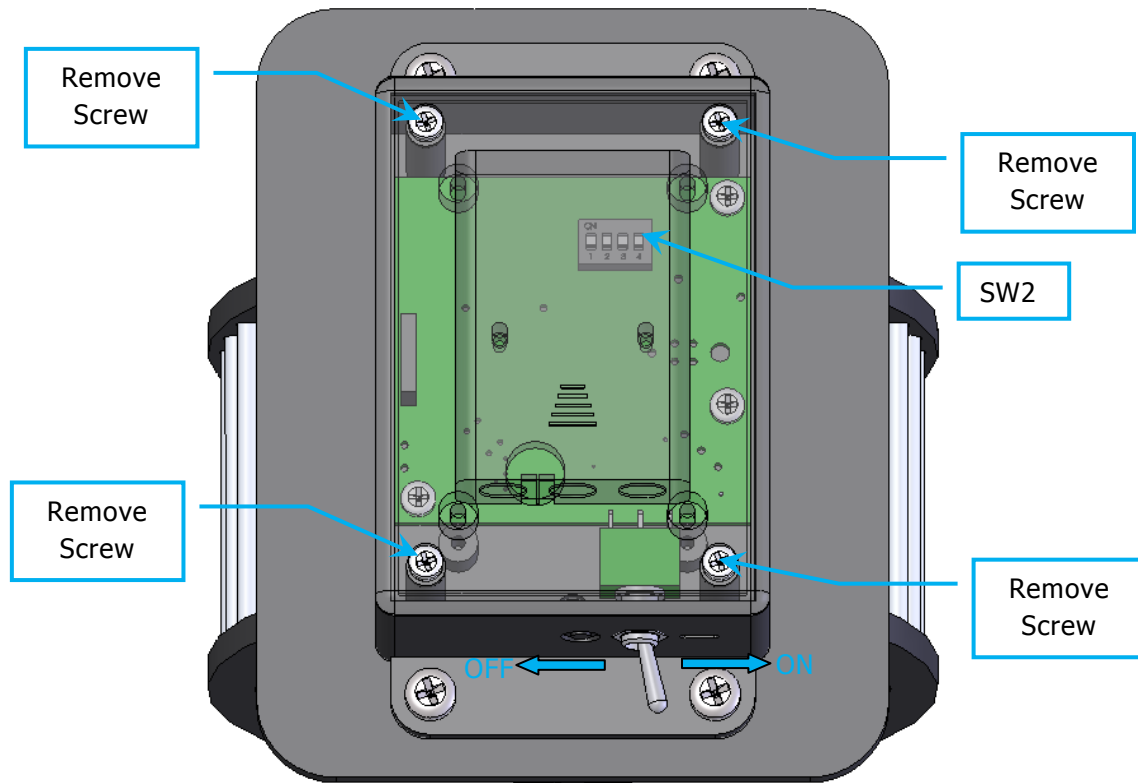
The channel number of the DIG-804 USB Interface Hub must be the same as the channel number of all of the ENV-044V Wireless Vertical Running Wheels that it will be monitoring. The channel number is set using switches on the devices as illustrated in Figure 3.1.

Figure 3.1 – Dip Switch Setting Example



The DIG-804 switches are located on the back panel of the hub. To access the switches in the ENV-044V, remove cover with the battery pack as shown in Figure 3.2. This will expose the circuit board inside. The switches are labeled SW2.

Figure 3.2 – Accessing the Dip Switches (SW2)



CHAPTER 4

Software

The ENV-044V Wireless Vertical Running Wheel is designed to be used with SOF-860 Wheel Manager Software. Please refer to the Wheel Manager Software User's Manual for instructions on using the application.

CHAPTER 5

Cleaning Instructions

The top of the ENV-044V Running Wheel contains batteries and sensitive electronic components. It should be handled with care and should not be exposed to extreme temperatures. Never submerge the unit in any liquid. The exterior of the unit should be cleaned by hand using a mild, non-abrasive detergent such as Quatricide®.

The wheel can be washed in cage washers or lab dishwashers achieving temperatures up to 180° F.

Appendix A

Contact Information

Please contact MED Associates, Inc. for information regarding any of our products.

Visit our website at www.med-associates.com for contact information.

For technical questions, email support@med-associates.com.